SUMIT PATHAK

Pune, India | sumitpathak2904@gmail.com | 8076221099 | linkedin.com/in/sumitpathak29 | github.com/Sumitpathak29

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student (Symbiosis Institute of Technology, Pune) who builds and ships full-stack and LLM-integrated systems end-to-end — from fine-tuning and deploying a domain-specific LLaMA 3.1 8B QA system to implementing a decoder-only GPT architecture from scratch. Comfortable across the stack: React/Node/Express on the frontend and API layer, PyTorch and RAG pipelines on the ML side, and AWS for deployment. Looking to bring that range to a software engineering or ML engineering role.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, SQL, HTML, CSS
- **Frontend:** React.js, HTML5, CSS3, Responsive Web Design
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
- **Databases:** PostgreSQL, MongoDB, ChromaDB
- **AI / ML:** PyTorch, NumPy, Pandas, Scikit-learn, LLaMA 3.1, Llama 3.2, LoRA, QLoRA, RAG, Prompt Engineering, Transformers, LangChain, Ollama, Hugging Face (Transformers, PEFT, TRL)
- **Cloud & Tools:** AWS EC2, Git/GitHub, Jupyter, Gradio, Webpack, DFX

PROJECTS

Medical Research Assistant *Python, LangChain, ChromaDB, Llama 3.2, Ollama, Gradio, Hugging Face*

- Built a multi-document Retrieval-Augmented Generation (RAG) system that processes medical PDFs, generates semantic embeddings using Sentence Transformers, and stores document vectors in a persistent ChromaDB knowledge base.
- Implemented the RAG pipeline from scratch covering PDF extraction, overlapping text chunking, embedding generation, semantic retrieval, prompt construction, and grounded LLM generation using Llama 3.2 through Ollama.
- Extended the system with LangChain for document loading, text splitting, embeddings, vector-store integration, and retrieval, and developed a Gradio interface supporting multiple PDF uploads and source/page attribution.
- Experimented with parameter-efficient fine-tuning using LoRA/QLoRA, 4-bit quantization, Hugging Face Transformers, PEFT, and TRL to explore domain adaptation of Llama-based models.

GPT Language Model from Scratch *Python, PyTorch, NumPy*

- Implemented a character-level, decoder-only GPT architecture from scratch using PyTorch, including token and positional embeddings, multi-head causal self-attention, feed-forward networks, LayerNorm, and residual connections.
- Built the complete autoregressive language-model training pipeline, including character-level tokenization, mini-batch generation, cross-entropy loss, backpropagation, optimization, and next-token generation using the Tiny Shakespeare dataset.
- Implemented causal attention masking to prevent future-token information leakage and generated character-level text through autoregressive inference.

Task Manager Web Application *React.js, Node.js, Express.js, MongoDB, JWT*

- Built a full-stack task management app with JWT-based authentication/authorization protecting all API endpoints, supporting [N] concurrent users in testing.
- Designed and implemented REST APIs for task CRUD, prioritization, deadlines, filtering, and search, covering [N] endpoints total.
- Built a responsive React.js frontend integrated with the Express/MongoDB backend, and structured the codebase into modular routes, middleware, and controllers for maintainability.

OTHER PROJECTS

DBank — Decentralized Banking Application *Motoko, Internet Computer Protocol, JavaScript, Webpack* — built a decentralized banking app on ICP using Motoko canisters for deposits, withdrawals, and compound-interest logic, with a JavaScript frontend for real-time balance updates.

CERTIFICATIONS

- Nutanix Hybrid Cloud Fundamentals (NHCF) — March 2026
- Full-Stack Web Development Bootcamp — Node.js, Express.js, React.js, Databases, REST APIs, Deployment
- AI Engineering — LLM, RAG, QLoRA, Agents

EDUCATION

Bachelor of Technology (B.Tech), Computer Science 2023 – 2027
Symbiosis Institute of Technology, Pune